Alan Roy

alanyenroy@gmail.com | alanroy2@illinois.edu | (425) 295 - 4095

Education

University of Illinois Urbana-Champaign *Expected Graduation:*
BS in Mechanical Engineering | Minor in Electrical Engineering *December 2028*

- GPA: 3.98
- **Relevant Coursework:** Non-Linear Dynamics and Vibrations, Control Systems, Heat Transfer, Fluid Dynamics, Mechanical Design, Solid Mechanics

Leadership & Team Experience

iRobotics *Champaign, Illinois*
Autonomous Karting Series Captain, Weapon Lead, Mechanical Lead *2024 to Present*

- Started new team of 30 people working to create an autonomously driving electric kart
- Oversees system-integration of SLAM and telemetry processes with steer-by-wire and brake-by-wire systems
- Designed and built FSH, a 30lb vertical spinner combat robot which was semifinalist at National Havoc Robot League
- Manufactured parts on water-jet, manual lathe and mill
- Designed vertical spinner weapon stack to sustain speeds of 6000 RPM, surviving 10 fights without failure
- Designing high-torque clutch to handle large impacts for hammer-saw robot design
- Developed mechanical workshops to teach students about topics such as 3D printing, designing for manufacturing, material selection, and designing for assembly
- Working on iRobotics executive committee to organize and smooth operation within RSO

Projects

One Pound Combat Robot *Ongoing*

- Built and competed wide-body horizontal spinner robot at Robot Day and Engineering Open House competitions
- Designing new suplex-inspired vertical spinner with inverted self-righting system

Quadcopter Control System *Ongoing*

- Created custom 3.5 inch quadcopter frame to test flight code
- Coding custom PID and control system for flight in C++ built off Ardupilot

Four Leg Walking Robot *2026*

- Designed one degree of freedom driven walking robot to traverse variety of uneven terrain
- Utilizes custom designed bevel and helical gears to transfer torque and synchronize linkages

Job Experience

Cashier and Cook *Issaquah, WA*
Bento Ya Goemon *July 2022 to June 2025*

- Maintained organization when working on many concurrent tasks, including cooking, plating, and processing sales
- Interacted smoothly with customers and coworkers while working in hectic and busy environment

Skills

Software: Solidworks, Fusion 360, PTC Creo, Onshape, Python, Matlab, C++, ANSYS

Fabrication: 3D Printing (FDM/SLA), CNC Mill, CNC Lathe, Turning, Milling, Laser Cutting, Water Jetting

Other: GD&T, Google Workspace, Microsoft Office, Adobe Suite, Blender, LaTeX